

# Maritime Passenger Data Insights with SQL

## 1) INTRODUCTION:

This project aims to analyse Maritime passenger data contained in multiple CSV files(passengers.csv, tickets.csv, survivals.csv, families.csv, and embarkation.csv) using SQL queries. The goal is to derive insights into passenger demographics, survival rates, and various factors contributing to survival. This analysis helps to understand how different attributes such as class, gender, age, family size, and embarkation port affected survival rates.

## 2) DATASET OVERVIEW:

### 2.1) **Dataset Name:** Ship Passenger Dataset

The dataset consists of five CSV files that contain information about Titanic passengers, their ticket details, survival status, family members, and embarkation points.

### 2.2) Attributes:

| COLUMN NAME   | DESCRIPTION                            |
|---------------|----------------------------------------|
| passenger_id  | Unique identifier for each passenger   |
| name          | Name of the passenger                  |
| gender        | Gender of the passenger (Male/Female)  |
| age           | Age of the passenger                   |
| ticket_id     | Unique ticket identifier               |
| ticket_number | Ticket number                          |
| class         | Class of the passenger (1st, 2nd, 3rd) |

|                                 |                                                                      |
|---------------------------------|----------------------------------------------------------------------|
| <b>fare</b>                     | Fare paid by the passenger                                           |
| <b>survived</b>                 | Whether the passenger survived (1 = Yes, 0 = No)                     |
| <b>siblings_spouses_ aboard</b> | Number of siblings/spouses aboard with the passenger                 |
| <b>parents_children_ aboard</b> | Number of parents/children aboard with the passenger                 |
| <b>embarked</b>                 | Port of embarkation (C = Cherbourg, Q = Queenstown, S = Southampton) |

### 2.3) Sample Data:

Here's a preview of the dataset:

#### **passengers.csv:**

Total Rows: 20

Total Columns: 4 (passenger\_id, name, gender, age)

Unique Genders: Male, Female

Age Range: 16 to 50 years

| Passenger_id | name          | gender | Age |
|--------------|---------------|--------|-----|
| 1            | John Smith    | Male   | 35  |
| 2            | Jane Doe      | female | 28  |
| 3            | Alice Johnson | female | 22  |

#### **tickets.csv:**

Total Rows: 20

Total Columns: 5 (ticket\_id, passenger\_id, ticket\_number, class, fare)

Ticket Classes: 1st, 2nd, 3<sup>rd</sup>

| Ticket_id | Passenger_id | Ticket_number | class           | Fare |
|-----------|--------------|---------------|-----------------|------|
| 101       | 1            | TCKT0001      | 1 <sup>st</sup> | 100  |
| 102       | 2            | TCKT0002      | 2 <sup>nd</sup> | 105  |
| 103       | 3            | TCKT0003      | 3 <sup>rd</sup> | 110  |

### **survivals.csv:**

Total Rows: 20

Total Columns: 2 (passenger\_id, survived)

Survival Status: 1 (Survived), 0 (Did Not Survive)

Survival Rate: Varies based on class, gender, and other factors.

| Passenger_id | Survived |
|--------------|----------|
| 1            | 1        |
| 2            | 0        |
| 3            | 1        |

### **families.csv:**

Total Rows: 20

Total Columns: 3 (passenger\_id, siblings\_spouses\_aborad, parents\_children\_aborad)

| Passenger_id | siblings_spouses_aborad | siblings_spouses_aborad |
|--------------|-------------------------|-------------------------|
| 1            | 1                       | 0                       |
| 2            | 0                       | 1                       |
| 3            | 0                       | 1                       |

### embarkation.csv:

Total Rows: 20

Total Columns: 2 (passenger\_id, embarked)

Embarkation Points: C (Cherbourg), Q (Queenstown), S (Southampton)

| Passenger_id | Embarked |
|--------------|----------|
| 1            | S        |
| 2            | C        |
| 3            | Q        |

### 3) SQL OBJECTIVES:

The following objectives were achieved during this SQL project:

- a) Mastering SQL JOINS (INNER JOIN, LEFT JOIN) to combine data from multiple tables.
- b) Aggregating data to summarize numerical information like survival rates, family size, and fare distribution.
- c) Analysing survival rates based on various attributes such as class, gender, and embarkation point.

### 4) METHODOLOGY USED:

| CONCEPT     | SQL COMMAND/FUNCTION             |
|-------------|----------------------------------|
| Filtering   | WHERE, IN, NOT IN, BETWEEN, CASE |
| Aggregation | SUM, AVG, COUNT, ROUND           |
| Joins       | INNER JOIN, LEFT JOIN            |

|          |                  |
|----------|------------------|
| Sorting  | ORDER BY, LIMIT  |
| Grouping | GROUP BY, HAVING |

## 5) DATA ANALYSIS AND KPIs:

1. What percentage of passengers survived overall?

**Query:**

```
SELECT ROUND(AVG(survived)*100,2) AS survival_rate_percent
FROM survivals;
```

**Insight:**

The overall survival rate is 50%

2. What was the survival rate by passenger class?

**Query:**

```
SELECT t.class, ROUND(AVG(s.survived) * 100, 2) AS
survival_rate_percent
FROM tickets t
JOIN survivals s ON t.passenger_id = s.passenger_id
GROUP BY t.class;
```

**Insight:**

2nd class had the highest survival rate (71.43%), followed by 1st (42.86%) and 3rd (33.33%).

3. How did survival rate differ between genders in each class?

**Query:**

```
SELECT t.class, p.gender, ROUND(AVG(s.survived) * 100, 2) AS
survival_rate_percent
FROM tickets t
```

```
JOIN survivals s ON t.passenger_id = s.passenger_id
JOIN passengers p ON t.passenger_id = p.passenger_id
GROUP BY t.class, p.gender;
```

**Insight:**

2nd class males had 100% survival, while 1st class females had 0% - very unusual trend here. Generally, survival wasn't uniformly distributed across gender & class.

4. What is the average family size for survivors vs. non-survivors?

**Query:**

```
SELECT s.survived, ROUND(AVG(f.siblings_spouses_aborad +
f.parents_children_aborad), 2) AS avg_family_size
FROM survivals s
JOIN families f
ON s.passenger_id = f.passenger_id
GROUP BY s.survived;
```

**Insight:**

Survivors had an average family size of 1.30, while non-survivors had 1.40 -very slight difference. Family size didn't drastically influence survival.

5. Which embarkation point had the most passengers?

**Query:**

```
SELECT e.embarked, COUNT(*) AS total_passengers
FROM embarkation e
GROUP BY e.embarked
ORDER BY total_passengers DESC;
```

**Insight:**

Most passengers boarded from Southampton (S) with 10 passengers, twice as many as Cherbourg (C) and Queenstown (Q).

6. What was the survival rate by embarkation port?

**Query:**

```
SELECT e.embarked, ROUND(AVG(s.survived) * 100, 2) AS  
survival_rate_percent  
FROM embarkation e  
JOIN survivals s ON e.passenger_id = s.passenger_id  
GROUP BY e.embarked;
```

**Insight:**

Passengers from Queenstown (Q) had the highest survival rate at 60%, followed by Southampton (50%) and Cherbourg (40%).

7. Identify passengers who did not survive and analyze their family size.

**Query:**

```
SELECT p.name, f.siblings_spouses_ aboard +  
f.parents_children_ aboard AS family_size  
FROM passengers p  
LEFT JOIN survivals s ON p.passenger_id = s.passenger_id  
LEFT JOIN families f ON p.passenger_id = f.passenger_id  
WHERE s.survived = 0;
```

**Insight:**

Passengers who did not survive often had small-to-moderate family sizes. 3 passengers had 2 family members, indicating that even with company, survival wasn't guaranteed.

8. Did people who paid more survive more often?

**Query:**

```
SELECT CASE
  WHEN t.fare >= 70 THEN 'High'
  WHEN t.fare BETWEEN 30 AND 70 THEN 'Medium'
  ELSE 'Low'
END AS fare_category,
ROUND(AVG(s.survived) * 100, 2) AS survival_rate_percent
FROM tickets t
JOIN survivals s ON t.passenger_id = s.passenger_id
GROUP BY fare_category;
```

**Insight:**

Passengers who paid higher fares had a 50% survival rate, same as the overall average - fare amount didn't guarantee better survival odds here.

9. Who were the top 5 oldest survivors?

**Query:**

```
SELECT p.name, p.age, t.class, p.gender
FROM passengers p
JOIN survivals s ON p.passenger_id = s.passenger_id
JOIN tickets t ON t.passenger_id = p.passenger_id
WHERE s.survived = 1
ORDER BY p.age DESC
LIMIT 5;
```

**Insight:**

The oldest survivor was George Clooney (50) from 3rd class. Most of the top survivors were males from 1st class, suggesting age wasn't a limiting factor for survival.

## 10. Determine gender-wise survival rate in each class

### Query:

```
SELECT t.class, p.gender, ROUND(AVG(s.survived) * 100, 2) AS  
survival_rate  
FROM passengers p  
JOIN survivals s ON p.passenger_id = s.passenger_id  
JOIN tickets t ON p.passenger_id = t.passenger_id  
GROUP BY t.class, p.gender;
```

### Insight:

Males in 2nd class (100%) had the best survival outcome. 1st class females (0%) had the worst. This shows a non-uniform survival pattern - not just about gender or class, but the combo mattered.

## 6) CONCLUSION:

This analysis provided valuable insights into survival patterns among passengers aboard a ship, based on factors such as class, gender, fare, family size, and embarkation point. The study revealed that survival rates varied significantly across different groups, with certain classes and genders showing higher chances of survival. Meanwhile, elements like fare and family size had a more limited impact. These findings emphasize the importance of data-driven planning for safety protocols and offer a solid foundation for deeper analysis in future maritime datasets.